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Enhanced electromagnetic wave coupling in infrared range at graphene-loaded $LiNbO_3$ interface

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ABSTRACT

The integration of 2D materials with photonic device substrates represents a rapidly evolving area of research within the field of all-optical devices. Combining $LiNbO_3$ (LN) with 2D materials holds particular promise for the development of novel photonic devices at micro- and nanostructural levels. In this study, we explore the surface electrical and optical properties of submicron-thick LN thin films loaded with monolayer graphene using Kelvin probe force microscopy (KPFM) and prism-coupled total reflection methods. Experimental findings reveal that while x-cut LN substrates exhibit no polarization effect, the introduction of graphene induces a potential difference due to atomic reconfiguration at the LN surface. This, in turn, influences the carrier concentration (n_s) in graphene. The graphene-loaded LN film demonstrates robust coupling to near-surface waves under infrared light excitation. This enhanced coupling may stem from surface plasmon polariton (SPP) excitations of graphene, facilitated by the heightened n_s on x-cut LN substrates. This coupling enhancement not only optimizes the performance of LN-based photonic devices and facilitates the miniaturization and integration of the devices, but also opens up the possibility of realizing the manipulation of photon transmission using waveguide arrays or photonic crystals based on the enhanced evanescent coupling.

1. Introduction

As an artificially synthesized optical crystal renowned for its remarkable versatility, LN finds extensive applications in optical and optoelectronic functional devices due to its exceptional ferroelectric, piezoelectric, photoelectric, and nonlinear optical attributes. Traditionally, optoelectronic devices employing LN, such as modulators, have relied on bulk LN crystals [1]. However, as device applications continue to shrink in size and integration requirements surge, there has been a consistent trend toward downsizing LN materials, exemplified by the emergence of LN-on-insulator (LNOI) [2–4] materials. The surface of LN or the interface between upper-layer material and the LN substrate holds paramount importance for the performance of LNOI devices, especially with the introduction of 2D materials on submicron LN surfaces [5–8]. Serving as a substrate for 2D materials, LNOI facilitates modulation of the properties of 2D materials and expands the spectrum of interactions between the surrounding environment and the LN surface, thereby enhancing involved physical processes. For instance, device functionality in submicron thin-film materials such as LNOI primarily depends on its waveguide structure. Modulating transmitted light within the waveguide as waveguide dimensions drop can be accomplished by modifying the electromagnetic wave characteristics close to the waveguide surface, which can have a substantial effect on the functioning and performance of LNOI devices.

The integration of 2D materials with silicon-based photonic devices has garnered significant attention in the realm of all-optical devices, yielding notable research outcomes [9–13]. Particularly, the fusion of 2D graphene with silicon-based devices has emerged as a focal point. Studies have showcased the enhancement of performance in photonic devices owing to various excellent properties of graphene. The tunable Fermi level (E_F) and high carrier mobility enable graphene to have complex optical dispersion and optical absorption that can be quickly modulated. Liu et al. [14] controlled the E_F of monolayer graphene by employing an applied bias voltage to realize the transmission of light in graphene-based optical modulators. Yao et al. [15] demonstrated the gated intracavity tunability of graphene-based optical frequency combs by varying E_F to modulate their second- and higher-order chromatic dispersion. Meanwhile, by utilizing graphene's light absorption leading

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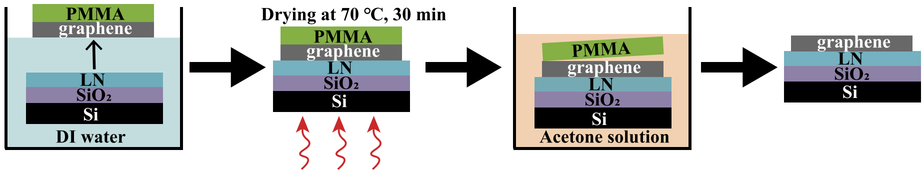

Fig. 1. Schematic illustration of PMMA-assisted wet transfer of graphene.

to nonlinear excitation, researchers achieved real-time detection of individual gas molecules in vacuum [16]. In addition, graphene's ultra-high thermal conductivity as well as low light absorption coefficient led to its first reported direct application in the field of graphene-silicon based thermo-optic modulators in 2013 [17]. In designing such devices, the interaction between the surface of substrate and graphene plays a pivotal role, often achieved by attaching graphene to thin-film waveguide structures.

Heterostructures combining graphene and LN have received scant research attention in the realm of optical devices, despite LN's pivotal role in optoelectronic devices. Currently, research on graphene-LN heterostructures primarily focuses on utilizing LN as a gate-control material for field-effect transistors, leveraging LN's interfacial property to modulate carriers in graphene via electric field effects or spontaneous polarization [18,19]. LN, being an anisotropic crystalline material, exhibits strong spontaneous polarization, significantly impacting physical and chemical properties near its surface [20–23]. This phenomenon is particularly pronounced in z-cut LN, where cutting LN perpendicular to the polarization axis results in the formation of a macroscopic electric dipole moment perpendicular to the surface.

In this work, we present the results of near-surface electrical and optical characterization for slab waveguide of 2D graphene on LNOI (graphene-LNOI, G-LNOI). Considering the influence of LN depolarization field on near-surface graphene, we opt for x-cut LN as the substrate for graphene. Choosing x-cut LN, the only LN surface exhibiting both nonpolar and nonpiezoelectric properties, minimizes the impact of physical and chemical processes introduced by the surrounding environment on the structural properties of G-LNOI.

2. Experiments and methods

2.1. Sample preparation

Composite waveguide structures of G-LNOI are fabricated by wet transferring CVD-grown monolayer graphene onto thin-film LNs (300 nm and 600 nm) on SiO_2/Si (2 μm/500 μm) substrates. A 9 mm × 9 mm monolayer graphene sheet is released in deionized water (DI water) and subsequently transferred onto a polished x-cut LN surface measuring 10 mm × 10 mm in area. Following removal from DI water, the samples are allowed to air dry at room temperature for 20 min before being placed in a 70 °C oven for 30 min to complete the drying process. Subsequently, the dried samples undergo soaking in an acetone solution to ensure complete removal of poly (methyl methacrylate) (PMMA) from the surface, resulting in graphene/LN/SiO_2/Si composite waveguide structures. The sample preparation process is illustrated in Fig. 1.

2.2. Characterization

KPFM is employed based on the Kelvin method [24] to measure the potential difference between the probe and sample surfaces, offering nanoscale surface potential quantification with high precision [25,26]. This non-destructive testing method is particularly suitable for measuring the potential difference between the surface of monolayer graphene and the LN substrate in our experiments. The amplitude-modulated (AM)-KPFM involves scanning the same sample twice to determine both surface morphology and surface potential. The tests are conducted at room temperature, with the probe positioned at one edge of the monolayer graphene, as depicted in Fig. 2(a).

To observe the effect of surface graphene on LNOI waveguides, we utilize the prism-coupled total reflection method based on total internal reflection or Otto configuration, where a rutile prism is placed in proximity to the surface of G-LNOI. Light traverses through the air gap between the prism and G-LNOI surface. When total reflection occurs, the evanescent wave propagates laterally along the graphene/LN interface, interacting with the graphene layer. This interaction affects the excitation of guiding modes in the LN waveguide, and the resultant impact is reflected in the reflectance spectra measured by prism coupling.

3. Results and discussions

Atomic force microscopy (AFM) and AM-KPFM techniques are utilized to investigate the morphology and surface potential distribution of graphene transferred onto 300 nm and 600 nm LN surfaces, as depicted in Fig. 2(b)–(e). Additionally, simultaneous tests are conducted on LNOI to mitigate the influence of the surrounding environment in subsequent analytical comparisons as shown in Fig. S1. Fig. 2(b) displays the surface morphology of 300 nm LN covered with graphene, revealing a distinct boundary between the graphene layer and the smooth LN surface. Along the path of line 1 in Fig. 2(b), a noticeable bulge at the edge caused by graphene wrinkles is observed, accompanied by a relatively uniform step-like jump on both sides of the graphene boundary. The average height difference between the two sides of the boundary is measured at 1.5 nm, representing the average thickness of the monolayer graphene wet transferred onto the LN surface. Fig. 2(c) illustrates the surface potential measurements for the same sample region as in Fig. 2(b), along the path of line 2. The results indicate a lower potential in the graphene-covered region compared to the LN surface, with a potential difference of approximately 32 mV between them.

Surface test results for the 600 nm LN film sample with a single layer of graphene transferred are shown in Fig. 2(d). The test, conducted along the path of line 3, reveals an average height difference of only 1.136 nm between the LN surface and the uniformly distributed graphene. Despite an evident bump at the graphene boundary due to wrinkles from wet transfer, this suggests that most of the graphene on the LN exhibits the structure of a single layer. The subsequent surface potential test along the path of line 4 in the same region is presented in Fig. 2(e). Compared with Fig. 2(c), the potential difference between graphene and LN significantly increases to approximately 79.04 mV. These results suggest that graphene on LN accumulates negative charges relative to the x-cut LN surface, with the LN substrate thickness playing a pivotal role in influencing the electron concentration in graphene.

KPFM results reveal a step potential difference between the x-cut LN surface and monolayer graphene, similar to the voltage difference observed between graphene loaded on z-cut periodically poled LN, attributed to the modulation of graphene n_s by different LN substrate polarization directions [27,28]. Our experimental results demonstrate that graphene also exhibits negative charge accumulation relative to the x-cut LN surface. The x-cut surface of LN, typically referred to as the atomic structure section terminating perpendicular to LN $[2\,\bar{1}\,\bar{1}\,0]$

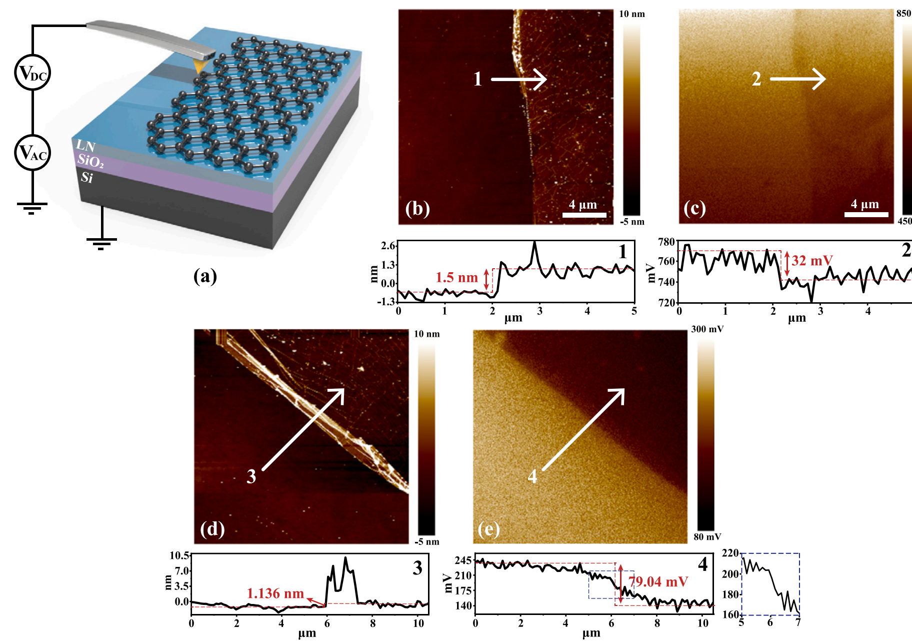

Fig. 2. (a) Three-dimensional schematic diagram of AM-KPFM setup on G-LNOI structure. The AFM topographic image and the corresponding AM-KPFM image for a monolayer graphene on (b)–(c) 300 nm LN and (d)–(e) 600 nm LN, respectively. The illustration below Figure (b)–(e) shows the height or potential profile along the white line in corresponding figure, with the *x*-axis direction corresponding to the direction indicated by the arrows. The blue dashed box on the right of the inset of (e) is the potential difference at the boundary between LNOI and the stacked graphene with modified *y*-axis span. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

direction, exhibits various termination possibilities. However, theoretical and experimental comparisons suggest that the -Li_{12} terminated (2 $\overline{1}$ $\overline{1}$ 0) surface predominates for x-cut LN [29–31]. The x-cut surface, theoretically unpolarized and non-piezoelectric, undergoes stabilization through processes such as adsorption of negatively charged oxygen atom groups (-O_3) [32] onto it or movement of surface cations into the crystal. The stabilized -Li_{12} terminated (2 $\overline{1}$ $\overline{1}$ 0) interface produces an electric dipole moment on the x-cut surface, pointing toward the crystal interior.

This analysis suggests that when monolayer graphene is loaded onto the x-cut LN surface, surface relaxation processes or the presence of electric dipoles affect the n_s in graphene, combined with physical adsorption on the surface, thereby achieving a form of “doping” of graphene to some extent. Compared to graphene on z-cut LN surfaces, where the effect of the substrate on n_s mainly stems from physical adsorption, the voltage difference between graphene and x-cut LN is smaller, indicating a somewhat lower n_s. However, x-cut LN surface relaxation appears to play a primary role in influencing graphene n_s, with enhanced surface relaxation and charge passivation occurring with increased total ferroelectric domain structure in the LN film. This is evidenced by the increase in potential difference between graphene and LN as the thickness of the x-cut LN film increases. For example, when the thickness of x-cut LN film is increased from 300 nm to 600 nm, the surface potential difference between graphene and LN is increased from 30 mV to about 80 mV, and the potential difference of 80 mV is close to that between graphene and z-cut LN substrate (0.1 V~0.3 V).[27]

Laser microscopic Raman spectroscopy serves as an effective method for non-destructive testing of n_s in graphene [33]. To confirm the effect of LN substrate on graphene n_s, Raman spectroscopy measurements are conducted for graphene on 300 nm and 600 nm LN substrates, with monolayer graphene transferred to SiO_2 substrate serving as a control group. Fig. 3(a) presents the Raman spectra obtained from the testing of three samples under excitation by a 488 nm laser with a power of 0.7 mW. The characteristic peaks corresponding to graphene are clearly visible in all three samples, with the 2D peak considered the most important feature for characterizing the number of graphene layers and being insensitive to doping. The stabilization of the peak frequency at approximately 2700 cm^{-1} confirms the monolayer properties of graphene on all three substrates [34].

The change in n_s of graphene is mainly reflected in the Raman spectra in terms of the frequency position of the G band, the full width at half-maximum (FWHM), and the ratio of I_{2D}/I_G. Results show a significant frequency shift of the G band in graphene on 300 nm (1583 cm^{-1}) and 600 nm (1594 cm^{-1}) LN substrates, respectively, with a G peak frequency shift of up to 11 cm^{-1}, along with a reduction in FWHM of the G peak. Similarly, the decrease in $\alpha = I_{2D}/I_G$ ratio with increasing n_s is experimentally verified ($\alpha_1>\alpha_2>\alpha_3$). This suggests that LN surface atomic reconfiguration and the work function difference between the LN substrate and graphene contribute to the doping effect on graphene, with the 11 cm^{-1} frequency shift of the G peak representing a change in n_s much greater than 4×10^{12} cm^{-2}[35].

To investigate the effect of graphene incorporation on LNOI waveguide properties, we examine the reflection spectra of G-LNOI using prism-coupled total reflection method, as depicted in the insert of Fig. 4(a). Samples are excited by a laser beam at wavelengths of 633 nm and 1539 nm, respectively, and the relationships between reflected intensity and effective mode refractive index (n_{eff}) in TE polarization mode are analyzed. The results are obtained based on formula $n_{eff} = n_1 \sin\theta$, where n_1 is the refractive index of LN waveguide layer, θ is the angle between the incident light and the normal direction of prism. Fig. 4(a)–(b) show the test results of different LN layer thicknesses (300 nm, 600 nm) loaded with graphene at 633 nm wavelength. The black lines represent test results for LNOI without graphene loading, serving as control groups. And the position marked by the blue dashed line corresponds to the guiding mode n_{eff} for theoretically simulated LNOI waveguide mode (see the computational details in Supporting Information). Compared with the results of multiple experiments (as shown in Fig. S2), it is observed that in 300 nm and 600 nm LNs, the coupling strength and n_{eff} of the fundamental mode are minimally affected and irregularly changed by the addition of graphene. However, the increase in coupling efficiency of higher-order modes is linked to the enhancement of attenuation wave amplitude in waveguide

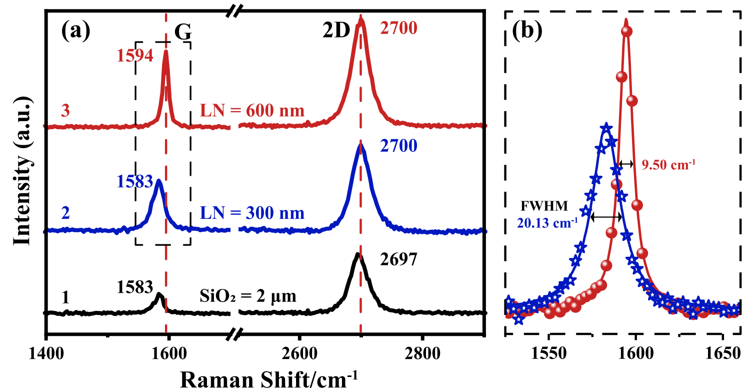

Fig. 3. (a) Raman spectra of the G and 2D peaks of three different graphene samples. (b) Enlarge and compare the G peak in the black dashed box of (a).

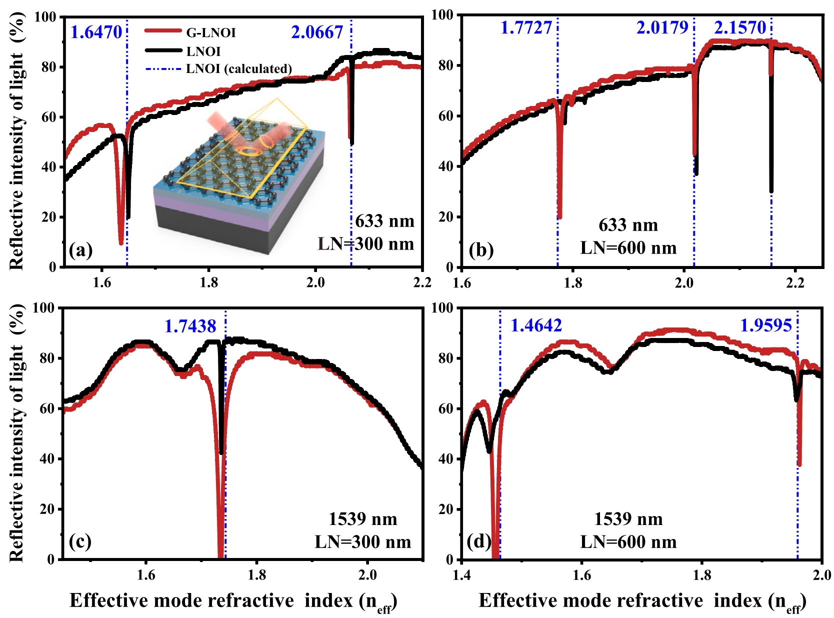

Fig. 4. Reflective spectra from LNOI and G-LNOI under TE polarization at the wavelength of (a)(b) 633 nm and (c)(d) 1539 nm. The insert in (a) is a schematic of the prism-coupled total reflection method. The blue dashed lines correspond to n_{eff} of the LNOI waveguide modes for each diagram calculated theoretically. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

cladding. In the visible wavelength band, graphene exhibits dielectric properties, with its incorporation having minimal effect on the transmission properties of LNOI waveguides.

The reflection spectrum under 1539 nm excitation displays a significantly larger decrease in reflection peaks of corresponding coupled modes compared to excitation at 633 nm. This reflects the enhancement of coupling efficiency between the LNOI waveguide mode and the electromagnetic wave near the surface under 1539 nm excitation. To eliminate any influence on test results due to changes in experimental conditions, tests are conducted on all samples under the same parameters. Unlike 633 nm excitation, where coupling efficiency varies little from mode to mode, repeated tests on samples under 1539 nm excitation consistently show a significant enhancement in prism coupling efficiency for G-LNOI. This result is clearly related to the nature of graphene at infrared wavelengths. Applications of graphene combined with waveguide structures in modulation and sensor devices have been widely reported, e.g., coupled plasma waveguide resonance and waveguide-coupled surface plasmon resonance (SPR) are two typical examples of utilizing the interfacial properties of waveguides and 2D graphene. Typical SPR requires a metal film as the excitation source of SPP, and the graphene introduced in SPR devices mainly serves as a dielectric film. Nowadays, many reports focus on the study of graphene SPR in the terahertz region [36,37], but its exploration in the near-infrared region is also indispensable [38,39]. It is found that the introduction of graphene can slightly increase the intensity of SPP and enhance the coupling between SPP and waveguide mode [40,41], which has become the most popular method for SPR devices to be coupled with waveguide modes and an important basis for introducing graphene in SPR devices to improve device performance.

Our experimental results show that the efficiency of prism coupling at an excitation wavelength of 1539 nm is significantly enhanced due to the incorporation of graphene, whereas there is no significant change in coupling efficiency at 633 nm excitation. This result is consistent with the different dielectric properties exhibited by graphene at the two wavelengths. However, for G-LNOI, its metallic properties are further enhanced due to the increase of its n_s. If the surface plasmon polaritons in graphene (GSPP) is excited, the coupling between it and waveguide mode greatly enhances the prism coupling efficiency, which is likely to be an important reason for the enhancement of prism coupling efficiency under 1539 nm excitation. Considering the unique properties of 2D graphene, there is an obvious difference between metallic graphene and metal in the excitation of SPP excitations. Studies on 2D graphene show that the excitation bandwidth of the GSPP is closely related to the chemical potential (μ_c) of graphene, which is proportional to n_s. It is

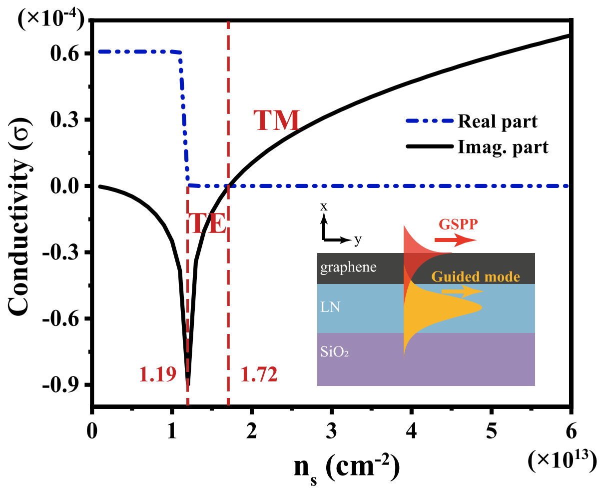

Fig. 5. Graphene conductivity as a function of n_s. The inset is a diagram of the coupling between GSPP and waveguide mode in the G-LNOI structure.

shown in the Kubo model [42] of graphene that the excitation of GSPP exhibits a robust relationship with electrical conductivity (σ).

$$\sigma = \sigma_{intra} + \sigma_{inter} \tag{1}$$

$$\sigma_{intra} = \frac{ie^2 k_b T}{\pi\hbar^2(\omega + i2\Gamma)}\left(\frac{\mu_c}{k_b T} + 2ln(e^{-\frac{\mu_c}{k_b T}} + 1)\right) \tag{2}$$

$$\sigma_{inter} = \frac{ie^2}{4\pi\hbar} ln\left(\frac{2\mu_c - (\omega + i2\Gamma)\hbar}{2\mu_c + (\omega + 2i\Gamma)\hbar}\right) \tag{3}$$

where $\mu_c \approx \hbar v_F \sqrt{n_s \pi}$ is set approximately equal to E_F. The scattering rate is defined as Γ=1/(2τ)=$ev_F^2/(2\mu E_F)$. For simplicity, we assume a constant fermi velocity $v_F = 1 \times 10^6$ m/s for intrinsic graphene in this work. Calculated from Eq. (1)-Eq. (3), the n_s range of TE GSPP mode excitation is 1.19×10^{13} cm^{-2}~ 1.72×10^{13} cm^{-2} at an incident wavelength of 1539 nm, while higher n_s can excite TM GSPP [43], and the results are shown in Fig. 5.

According to the analysis of n_s in graphene on x-cut LNOI substrate, the modulation range of n_s with different thicknesses of x-cut LNOI substrate is much larger than 4×10^{12} cm^{-2}, indicating the potential for graphene to reach an n_s capable of exciting GSPP. The enhancement of prism coupling efficiency under 1539 nm excitation may result from the coupling between GSPP and the guiding mode in the LNOI waveguide.

While further research and demonstration of GSPP excitation on G-LNOI are needed, the unique properties resulting from G-LNOI have opened up possibilities for designing and optimizing LN-based photonic devices. What is more, the addition of graphene strengthens the coupling between the waveguides via evanescent waves, which in a way serves to modulate special photonic lattice structures such as photonic Floquet lattices [44], which creates the possibility of manipulating photon transmission. For example, LN-based photonic crystal polarization beam splitters (Fig. 6(a)) may benefit from G-LNOI to increase the intensity of the transmitted modes in parallel waveguides while significantly shortening their lengths, thus optimizing and manipulating the transmission of light between waveguides.

Fig. 6(b) shows a directional beam splitter, the core component of polarization beam splitters, based on an isosceles trapezoidal shape LN ridge waveguide under actual preparation conditions for ion etching method, where the design parameters are W$_G$=660 nm, θ=75°, H=600 nm, W$_g$=100 nm, respectively. Fig. S4 reveals the comparative results of the coupling efficiencies of LNOI (0.9502) and G-LNOI

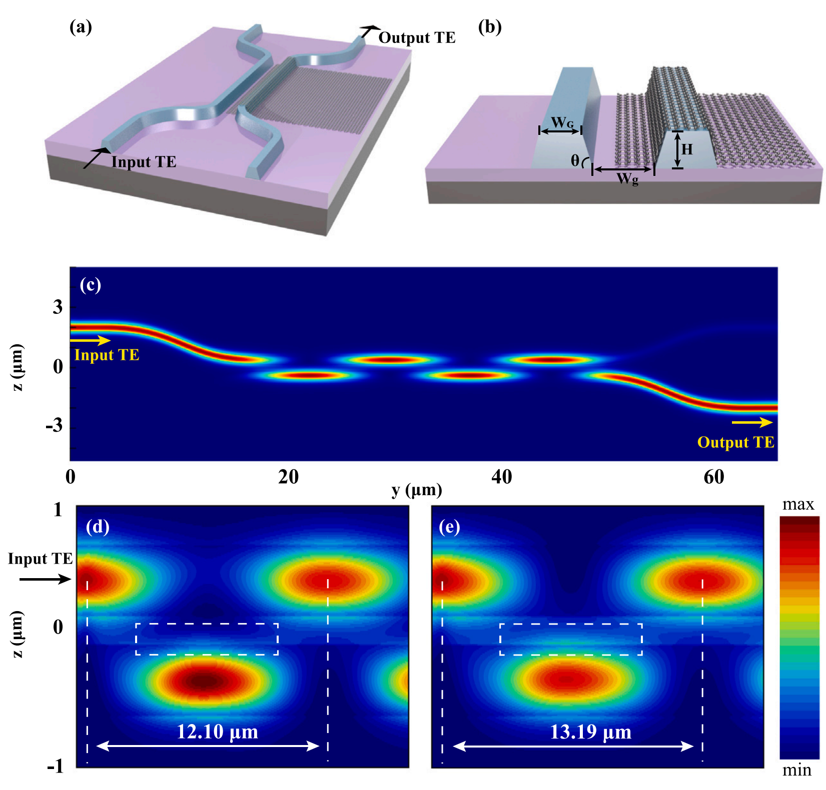

Fig. 6. Three-dimensional schematic diagram of (a) LN-based photonic crystal polarization beam splitter and (b) the part of directional coupler structure in (a). (c)Power maps of TE for polarization beam splitter. Power maps of TE for directional coupler with (d) G-LNOI and (e) LNOI, respectively.

(0.9630~0.9650) beam splitters based on our calculations of simulations. It can be seen that the addition of graphene can effectively improve the coupling efficiency between LN waveguides. The red star with 0.9650 coupling efficiency in the figure corresponds to the calculation result when n_s of graphene is 1.19×10^{13} cm^{-2}. Since the simulating calculation of the coupling efficiency includes the material and structural parameters of LN waveguide as well as the n_s of graphene, it is possible that the coupling efficiency can also be improved by optimizing the above parameters. The overall power map of the polarization beam splitter is shown in Fig. 6(c), which is simulated based on the finite-difference time-domain method. Simulation results show that the presence of graphene significantly affects the coupling period and intensity of waveguide at 1539 nm, with the optical power in G-LNOI stronger and the coupling period between two waveguides shorter (see Fig. 6(d)) compared to LNOI (Fig. 6(e)). The compressed coupling cycle can reduce the size of LN-based optical devices, improve device sensitivity and efficiency, and the strong coupling between the waveguides means that the path of optical transmission can be adjusted or altered. This result shows that the composite structure formed by loading graphene on LNOI films not only facilitates the miniaturization and integration of LN-based all-optical devices, but also offers the possibility of modulating photon transport using waveguide arrays or photonic lattices.

4. Conclusions

In summary, this investigation delves into the interfacial dynamics between graphene and x-cut LNOI, revealing enhanced coupling between near-surface electromagnetic waves and LN waveguide modes when infrared light excites the system. AM-KPFM experimental findings highlight a substantial potential difference, reaching tens of millivolts, between the LN surface and the monolayer graphene loaded onto it. Raman spectroscopy results indicate a correlation between the n_s in graphene and the thickness of the x-cut LN substrate, suggesting that LN surface atomic reconfiguration and the potential difference between LN and graphene contribute to "doping" graphene. Reflectance spectroscopy tests conducted on the G-LNOI composite structure reveal that the presence of graphene enhances coupling between LNOI waveguide modes and electromagnetic waves near the LN surface under infrared light excitation. This heightened coupling may be attributed to GSPP excitation. The combination of 2D graphene and LN thin-film waveguides for LN-based all-optical devices not only facilitates the miniaturization and integration of all-optical devices, but also offers the possibility of manipulating the transmission of photons, which provides a new avenue for the development of high-performance photonic devices and optical transmission systems in the future.

CRediT authorship contribution statement

Yifan Liu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Fei Lu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Hui Hu:** Supervision, Resources, Project administration. **Peiju Yin:** Resources, Investigation, Formal analysis. **Kai Zhao:** Resources, Investigation. **Yan Liu:** Visualization, Methodology. **Yao Wei:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.surfin.2024.104894.

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